INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Worksheet 2 Solutions

1. At each of two visits, a patient's blood pressure is classified as normal (N), elevated (E), or high (H). For example, NH means normal at the first visit and high at the second.

   (a) Write the sample space by listing its sample points.

   Solution: $S = \{\text{NN}, \text{NE}, \text{NH}, \text{EN}, \text{EE}, \text{EH}, \text{HN}, \text{HE}, \text{HH}\}$. By the Fundamental Principle of Counting, $|S| = 3 \cdot 3 = 9$.

   (b) Write the event A, "the classification is the same at both visits," by listing its sample points.

   Solution: $A = \{\text{NN}, \text{EE}, \text{HH}\}$.

   (c) Write the event B, "the blood pressure is high at least once," by listing its sample points.

   Solution: $B = \{\text{NH}, \text{EH}, \text{HN}, \text{HE}, \text{HH}\}$.

   (d) Find $A \cap B$ and $A \cup B$.

   Solution: $A \cap B = \{\text{HH}\}$ and $A \cup B = \{\text{NN}, \text{EE}, \text{HH}, \text{NH}, \text{EH}, \text{HN}, \text{HE}\}$.

2. After a certain outpatient surgery, patients stay in the hospital for 1 to 5 days, so the sample space for the length of a randomly chosen patient's stay is $S = \{1, 2, 3, 4, 5\}$. The probability of a stay of x days is

$$P(x) = \frac{6 - x}{15}.$$

(a) Write out $\sum_{x \in S} P(x)$ term by term, and check that it equals 1.

Solution:

$$\sum_{x \in S} P(x) = P(1) + P(2) + P(3) + P(4) + P(5) = \frac{5}{15} + \frac{4}{15} + \frac{3}{15} + \frac{2}{15} + \frac{1}{15} = \frac{15}{15} = 1.$$

(b) Let $A = \{3, 4, 5\}$ be the event "the patient stays at least 3 days." Write $P(A)$ using sigma notation, then evaluate it.

Solution: $P(A) = \sum_{x \in A} P(x) = P(3) + P(4) + P(5) = \frac{3}{15} + \frac{2}{15} + \frac{1}{15} = \frac{6}{15} = \frac{2}{5}$.

3. Two fair dice are rolled.

(a) What is the probability that the numbers the dice land on add up to at least 10?

Solution: Let A be the event "the numbers the dice land on add up to at least 10." Then

$$A = \{(4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)\},$$

so $|A| = 6$. The 36 sample points are equally likely, so

$$P(A) = \frac{|A|}{|S|} = \frac{6}{36} = \frac{1}{6}.$$

(b) What is the probability that the numbers the dice land on differ by 2?

Solution: Let B be the event "the numbers the dice land on differ by 2." Then

$$B = \{(1, 3), (2, 4), (3, 5), (4, 6), (3, 1), (4, 2), (5, 3), (6, 4)\},$$

so $|B| = 8$, and

$$P(B) = \frac{|B|}{|S|} = \frac{8}{36} = \frac{2}{9}.$$

4. In the blood type experiment, the blood types of the residents of a city are distributed as follows.

Blood type	A+	A−	B+	B−	AB+	AB−	O+	O−
Probability	0.32	0.08	0.08	0.02	0.04	0.01	0.36	0.09

Recall that a patient can safely receive red blood cells only from a donor whose red blood cells carry no antigen that the patient's own red blood cells lack. A resident of the city is chosen at random.

(a) What is the probability that the resident's red blood cells carry exactly one of the A and B antigens?

Solution: Let X be the event "the resident's red blood cells carry exactly one of the A and B antigens." Then

$$X = \{\text{A+}, \text{A−}, \text{B+}, \text{B−}\},$$

so

$$P(X) = P(\text{A+}) + P(\text{A−}) + P(\text{B+}) + P(\text{B−}) = 0.32 + 0.08 + 0.08 + 0.02 = 0.50.$$

(b) What is the probability that the resident's red blood cells carry at least one of the three antigens?

Solution: Let Y be the event "the resident's red blood cells carry at least one of the three antigens." The only blood type with none of the three antigens is O−, so $Y^{\text{c}} = \{\text{O−}\}$, and

$$P(Y) = 1 - P(Y^{\text{c}}) = 1 - P(\text{O−}) = 1 - 0.09 = 0.91.$$

(c) What is the probability that the resident can receive red blood cells from a type O+ donor?

Solution: Let Z be the event "the resident can receive red blood cells from a type O+ donor." A type O+ donor's red blood cells carry only the Rh antigen, so the resident can receive them exactly when the resident's own red blood cells carry the Rh antigen, that is, when the resident has a positive blood type. So

$$Z = \{\text{A+}, \text{B+}, \text{AB+}, \text{O+}\},$$

and

$$P(Z) = P(\text{A+}) + P(\text{B+}) + P(\text{AB+}) + P(\text{O+}) = 0.32 + 0.08 + 0.04 + 0.36 = 0.80.$$

5. At a clinic, 30% of the patients have asthma, 20% have seasonal allergies, and 42% have at least one of the two. A patient is chosen at random. Let A be the event "the patient has asthma" and L the event "the patient has seasonal allergies."

(a) Find $P(A \cap L)$.

Solution: By the Inclusion-Exclusion Principle for Events,

$$0.42 = P(A \cup L) = P(A) + P(L) - P(A \cap L) = 0.30 + 0.20 - P(A \cap L),$$

so $P(A \cap L) = 0.50 - 0.42 = 0.08$.

(b) What is the probability that the patient has exactly one of the two?

Solution: Of the 42% of patients who have at least one of the two, 8% have both, so the probability is $0.42 - 0.08 = 0.34$.

(c) Are A and L mutually exclusive? Are they independent? Explain.

Solution: A and L are not mutually exclusive, since $P(A \cap L) = 0.08 \neq 0$. They are not independent either, since $P(A) \cdot P(L) = 0.30 \cdot 0.20 = 0.06 \neq 0.08 = P(A \cap L)$.

6. A hospital has two backup generators, which work independently of each other. During a power outage, the first generator fails with probability 0.05, and the second fails with probability 0.08. Let F_1 be the event "the first generator fails" and F_2 the event "the second generator fails."

(a) Find $P(F_1 \cap F_2)$.

Solution: The generators work independently, so by the Multiplication Rule, $P(F_1 \cap F_2) = 0.05 \cdot 0.08 = 0.004$.

(b) What is the probability that at least one of the generators works?

Solution: The event "at least one of the generators works" is the complement of $F_1 \cap F_2$, so its probability is $1 - 0.004 = 0.996$.

(c) What is the probability that exactly one of the generators fails?

Solution: Either the first generator fails and the second works, with probability $0.05 \cdot 0.92 = 0.046$, or the first works and the second fails, with probability $0.95 \cdot 0.08 = 0.076$. These events are mutually exclusive, so by the Addition Rule the probability is $0.046 + 0.076 = 0.122$.

7. A hospital classifies its nurses by shift and by whether they have had a flu shot this season. The table gives the probabilities for a randomly chosen nurse.

	Has had a flu shot	Has not	Total
Day shift	0.42	0.18	0.60
Night shift	0.28	0.12	0.40
Total	0.70	0.30	1

Let D be the event "the nurse works the day shift" and F the event "the nurse has had a flu shot."

(a) Find $P(F \mid D)$ and $P(F \mid D^{\text{c}})$.

Solution: Knowing D shrinks the group to the first row, so $P(F \mid D) = \dfrac{0.42}{0.60} = 0.7$. Knowing D^{c} shrinks the group to the second row, so $P(F \mid D^{\text{c}}) = \dfrac{0.28}{0.40} = 0.7$.

(b) Are D and F independent? Explain.

Solution: Yes, since $P(F \mid D) = 0.7 = P(F)$. (Alternatively, $P(D) \cdot P(F) = 0.60 \cdot 0.70 = 0.42 = P(D \cap F)$.) Knowing a nurse's shift does not change the probability that the nurse has had a flu shot.

(c) Find $P(D^{\text{c}} \mid F^{\text{c}})$, and describe it in words.

Solution: Knowing F^{c} shrinks the group to the second column, so $P(D^{\text{c}} \mid F^{\text{c}}) = \dfrac{0.12}{0.30} = 0.4$. That is, 40% of the nurses who have not had a flu shot work the night shift.

(d) Find $P(D \cup F)$.

Solution: By the Inclusion-Exclusion Principle for Events, $P(D \cup F) = 0.60 + 0.70 - 0.42 = 0.88$.

8. A fair coin is tossed 6 times. Let E_k be the event "exactly k of the tosses land on heads."

(a) Find $P(E_k)$ for each k from 0 to 6.

Solution: This is a binomial experiment with $n = 6$ and $p = \frac{1}{2}$, so $P(E_k) = {}_6C_k \cdot \left(\frac{1}{2}\right)^6 = \frac{{}_6C_k}{64}$:

k	0	1	2	3	4	5	6
$P(E_k)$	$\frac{1}{64}$	$\frac{6}{64}$	$\frac{15}{64}$	$\frac{20}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$

(b) Write the sum of your seven answers to part (a) using sigma notation. Explain why this sum must equal 1, and check that it does.

Solution: The sum is $\sum_{k\in\{0,1,\ldots,6\}} P(E_k)$. Every sequence of 6 tosses has exactly one number of heads, so the events E_0, E_1, $\ldots$, E_6 are mutually exclusive and their union is all of S. By the Addition Rule, the sum must equal $P(S) = 1$. Indeed,

$$\frac{1+6+15+20+15+6+1}{64} = \frac{64}{64} = 1.$$

(c) What is the probability of getting more heads than tails?

Solution: More heads than tails means 4, 5, or 6 heads, so by the Addition Rule the probability is

$$P(E_4) + P(E_5) + P(E_6) = \frac{15+6+1}{64} = \frac{22}{64} = \frac{11}{32} \approx 0.344.$$

9. A practice quiz has 10 multiple-choice questions, and each question has 4 answer choices, exactly one of which is correct. A student guesses at random on every question. Let E_k be the event "exactly k of the student's answers are correct."

(a) Explain why this is a binomial experiment. What are n, a success, and p?

Solution: The number of questions, $n = 10$, is fixed, and each answer is either correct or not. The guesses are independent of each other, and the probability of guessing correctly is $p = \frac{1}{4}$ on every question. A success is a correct answer.

(b) Find $P(E_3)$.

Solution: $P(E_3) = {}_{10}C_3 \cdot 0.25^3 \cdot 0.75^7 = 120 \cdot 0.25^3 \cdot 0.75^7 \approx 0.250$.

(c) What is the probability that at least one of the student's answers is correct?

Solution: The complement is E_0, "no correct answers." Since $P(E_0) = 0.75^{10} \approx 0.056$, the probability is

$$1 - 0.75^{10} \approx 0.944.$$

(d) The student passes the quiz with at least 8 correct answers. What is the probability that the student passes by guessing?

Solution: The student passes exactly when E_8, E_9, or E_{10} occurs. These events are mutually exclusive, so by the Addition Rule the probability is

$$P(E_8) + P(E_9) + P(E_{10}) = {}_{10}C_8 \cdot 0.25^8 \cdot 0.75^2 + {}_{10}C_9 \cdot 0.25^9 \cdot 0.75 + 0.25^{10} \approx 0.0004.$$

Passing by guessing is extremely unlikely.

10. A box holds 20 rapid test kits, and 4 of them are defective. A nurse takes 3 kits from the box at random, one after the other, without putting any back.

(a) What is the probability that none of the 3 kits is defective?

Solution: Let G_1, G_2, and G_3 be the events that the first, second, and third kits are not defective. By the General Multiplication Rule,

$$P(G_1 \cap G_2 \cap G_3) = \frac{16}{20} \cdot \frac{15}{19} \cdot \frac{14}{18} = \frac{28}{57} \approx 0.491.$$

(b) What is the probability that at least one of the 3 kits is defective?

Solution: This is the complement of the event in part (a), so its probability is $1 - \frac{28}{57} = \frac{29}{57} \approx 0.509$.

(c) What is the probability that exactly one of the 3 kits is defective?

Solution: Write D for a defective kit and G for a good one. The event is

$$\{\text{DGG}, \text{GDG}, \text{GGD}\}.$$

By the General Multiplication Rule,

$$P(\text{DGG}) = \frac{4}{20} \cdot \frac{16}{19} \cdot \frac{15}{18} = \frac{8}{57},$$

and $P(\text{GDG})$ and $P(\text{GGD})$ have the same factors in a different order, so they are also $\frac{8}{57}$. These sample points are mutually exclusive, so by the Addition Rule the probability is $3 \cdot \frac{8}{57} = \frac{8}{19} \approx 0.421$. (Check by counting: $\frac{{}_4C_1 \cdot {}_{16}C_2}{{}_{20}C_3} = \frac{4 \cdot 120}{1{,}140} = \frac{8}{19}$.)

(d) A different clinic receives a very large shipment of kits, of which 20% are defective, so whether one kit is defective has no effect on whether another one is. A nurse there takes 3 kits at random. What is the probability that exactly one of them is defective? Compare with your answer to part (c).

Solution: Now the kits are independent, so this is a binomial experiment with $n = 3$ and $p = 0.2$, where a success is a defective kit. The probability of exactly one defective kit is

$${}_3C_1 \cdot 0.2 \cdot 0.8^2 = 3 \cdot 0.2 \cdot 0.64 = 0.384,$$

a little smaller than the answer to part (c).

11. Two unrelated residents of the city in Problem 4 are chosen at random, so their blood types are independent. We want to know how likely it is that the first resident can donate red blood cells to the second.

(a) Suppose the second resident has type O+ blood. What is the probability that the first resident can donate red blood cells to the second?

Solution: Let W be the event "the first resident can donate red blood cells to a type O+ patient." A type O+ patient's red blood cells carry only the Rh antigen, so the donor's red blood cells must carry no A or B antigen. So

$$W = \{\text{O+}, \text{O}-\},$$

and $P(W) = P(\text{O+}) + P(\text{O}-) = 0.36 + 0.09 = 0.45$.

(b) What is the probability that the second resident has type O+ blood and the first resident can donate red blood cells to the second?

Solution: The blood types are independent, so by the Multiplication Rule the probability is $P(\text{O+}) \cdot P(W) = 0.36 \cdot 0.45 = 0.162$.

(c) What is the probability that the first resident can donate red blood cells to the second?

Solution: We repeat part (b) for each blood type of the second resident. For each type, the first resident can donate when their blood type carries no antigen that the second resident's lacks:

Second resident	First resident can donate	Probability of both
A+	A+, A−, O+, O−: 0.85	$0.32 \cdot 0.85 = 0.272$
A−	A−, O−: 0.17	$0.08 \cdot 0.17 = 0.0136$
B+	B+, B−, O+, O−: 0.55	$0.08 \cdot 0.55 = 0.044$
B−	B−, O−: 0.11	$0.02 \cdot 0.11 = 0.0022$
AB+	any type: 1	$0.04 \cdot 1 = 0.04$
AB−	A−, B−, AB−, O−: 0.20	$0.01 \cdot 0.20 = 0.002$
O+	O+, O−: 0.45	$0.36 \cdot 0.45 = 0.162$
O−	O−: 0.09	$0.09 \cdot 0.09 = 0.0081$

These eight events are mutually exclusive, so by the Addition Rule the probability is

$$0.272 + 0.0136 + 0.044 + 0.0022 + 0.04 + 0.002 + 0.162 + 0.0081 = 0.5439.$$